

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Practicum 2: Exploring Probability: Uncertainty in the Mean
09/18/2025

In the following steps you will write a code that explores probability. Regardless of how you approach the problem, *you should use loops or array math to handle the calculations, rather than repeatedly writing out the same commands*. Remember to make screenshots at the end to upload to Webcourses for this practicum!

1. **(5 points)** Write a code that creates a subsample containing N random draws from a Gaussian distribution with a width $\sigma = 13$ and mean $\mu = 55$. You can use the help of AI to do this. What is the prompt you used?
2. **(5 points)** Try to do the following steps on your own, without the help of AI. Use your code to create a sample with $N=10$ random draws.
 - a. Do this 10 times (remember, don't repeat the same command!). For each sample, record the sample number (0 to 9), the sample mean, and the sample standard deviation in an array with one row per sample. Do not include the actual values of the samples in your array.

At the end you should have a 10 by 3 array with each line corresponding to one of your samples. For example,

```
2 62.35 3.25
```

would be the line for my sample 2, with a mean of 62.35 and a width of 3.25.

3. **(4 points)** Ask AI to do the same for you! What is the prompt you used? How does your code compare?
4. **(4 points)** Without the help of AI, use the appropriate Python commands to write the array into a text file. If you can't easily figure out how to write to a file, Google and Stackoverflow will help. Remember to name the output file following the class's formatting naming conventions.

In your output file, before the table, include a comment stating the number of draws in the table. Be careful to not overwrite your previous entries on the file when you access it to write a new result!

5. **(10 points)** Repeat (2a) for samples with $N=100$, 1,000, 10,000, 100,000, and 1,000,000 draws. Remember that you will need to do 10 repetitions for each of these sample sizes. Record each set of results in a separate 2D array and append them to the same file as in Problem 4.

6. **(10 points)** Initially, try to do the following on your own. If you are stuck use the help of AI. If you do use AI, what is the prompt you used?

Find the standard deviations of the means for each of the six sample sizes you ran. To do this, read in all the data you generated and stored. Use an appropriate Python function to find the standard deviation of your means for each sample size.

Then, print the sample size and its calculated standard deviation both on the screen and in your `.txt` (or `.dat`) file (be careful to not overwrite the data that is already there. Just append this info). For example, your output could look like this:

```
#Sample size    Std. dev. of mean
10              24.3
100             23.4
```

7. **(12 points)** Make a log-log plot of the standard deviation of the mean vs. sample size. The sample size should be the independent variable on your plot. Save as PNG using the appropriate Python command. Remember to name your plot appropriately.